

Madanapalle Institute of Technology and Science

SUBJECT: 23CAI103 – DIGITAL LOGIC AND COMPUTER ORGANIZATION

Assignment No: 2

Department: Computer Science and Engineering and Artificial Intelligence

Name: S Safa

Roll No: 24691A31M5

Section: CAI-D

Q1. Illustrate Restoring Division Algorithm with a suitable example. Divide $(22)_{10}$ by $(3)_{10}$

Ans:

Introduction:

The restoring division algorithm is a method used in computer organization to perform binary division. It works by repeatedly shifting and subtracting the divisor from the partial remainder. If the result becomes negative, the original value is restored.

Algorithm Steps

1. Initialize $A = 0$, $Q = \text{Dividend}$, $M = \text{Divisor}$

2. Repeat for n bits:

Left shift A and Q

Subtract M from A ($A = A - M$)

If $A < 0$, restore $A = A + M$ and set $Q_0 = 0$

Else set $Q_0 = 1$

3. Final Q gives quotient and A gives remainder

Example: $22 \div 3$

Decimal to Binary:

Dividend $(22)_{10} = 10110_2$

Divisor $(3)_{10} = 00011_2$

Initial Values:

$A = 00000$

$Q = 10110$

$M = 00011$

Iterations

Step 1: Shift $\rightarrow A=00001$, Subtract $\rightarrow$ Negative $\rightarrow$ Restore $\rightarrow Q_0=0$

Step 2: Shift $\rightarrow A=00010$, Subtract $\rightarrow$ Negative $\rightarrow$ Restore $\rightarrow Q_0=0$

Step 3: Shift $\rightarrow A=00101$, Subtract $\rightarrow$ Positive $\rightarrow Q_0=1$

Step 4: Shift $\rightarrow A=00010$, Subtract $\rightarrow$ Positive $\rightarrow Q_0=1$

Step 5: Shift $\rightarrow A=00101$, Subtract $\rightarrow$ Positive $\rightarrow Q_0=1$

Result

Quotient (Q) = $00111_2 = 7$

Remainder (A) = $00001_2 = 1$

Final Answer: $22 \div 3 = 7$ remainder 1

Q2. Explain in detail about the concept of Memory Hierarchy

Introduction

Memory hierarchy is the arrangement of different types of memory in a computer system based on speed, size, and cost. It helps achieve a balance between performance and storage capacity.

Levels of Memory Hierarchy

1. Registers:

Located inside CPU

Fastest memory

Very small storage

Used for immediate operations

2. Cache Memory:

Located between CPU and main memory

Stores frequently used data

Types: L1, L2, L3

Faster than RAM but smaller in size

3. Main Memory (RAM):

Stores currently executing programs

Volatile in nature

Moderate speed and size

4. Secondary Memory:

Permanent storage devices

Examples: Hard disk, SSD

Large storage capacity

Slower compared to primary memory

Characteristics

Speed decreases from top to bottom

Cost per bit decreases downward

Storage capacity increases downward

Advantages:

Improves system performance

Reduces cost

Provides large storage efficiently

Q3. Explain Flynn's Taxonomy of Parallel Processor Architectures

Introduction

Flynn's taxonomy is a classification of computer architectures based on the number of instruction streams and data streams processed simultaneously.

Types of Flynn's Taxonomy

1. SISD (Single Instruction Single Data)

One instruction and one data stream

Traditional sequential computers

Example: Single-core processor

2. SIMD (Single Instruction Multiple Data)

One instruction operates on multiple data

Used in vector processing and GPUs

3. MISD (Multiple Instruction Single Data)

Multiple instructions on a single data stream

Rarely used

Used in fault-tolerant systems

4. MIMD (Multiple Instruction Multiple Data)

Multiple instructions operate on multiple data streams

Used in modern parallel systems

Example: Multi-core processors

Conclusion:

Flynn's taxonomy helps in understanding different parallel processing architectures and their applications in modern computing system.